Before an AI System Answers Anything, a Language Model Knows How Well It Will Do

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Abstract

Given an AI system's graded answers to a few questions of a benchmark, we predict whether it answers each remaining question correctly. The hardest case is before it has answered any, because the predictor does not yet know how strong the system is. A language model does. We give it the system's name and a few of the benchmark's questions, ask for the system's accuracy, and use that as the starting probability that the system's answers then update. On the hidden benchmarks of the Predictive AI Evaluation Competition (PAIEC), this cut the error before the first answer by almost a quarter relative to guessing 50%, and it helped on 37 of 43 system–benchmark pairs. In simulation it is the largest gain in our predictor over the competition's full scoring. On public benchmarks it closes 78% of the gap between starting from a guess and starting from the system's true accuracy. Without the system's name it is barely better than a guess.

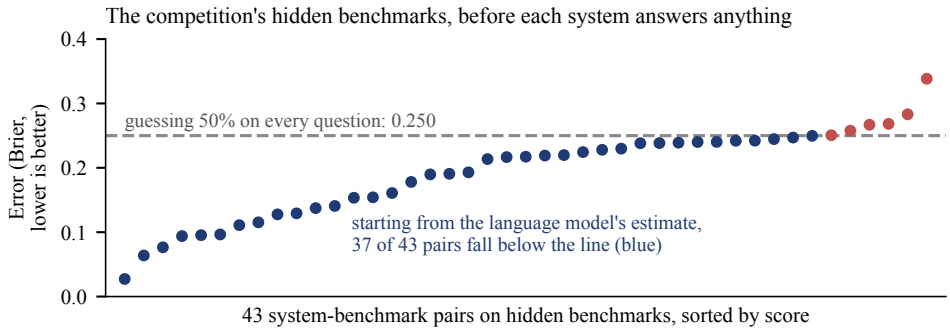


Figure 1: Error before the first answer on the competition's hidden benchmarks. Each dot is one system–benchmark pair from the five live runs that start from the language model's estimate. Without it the predictor answers one half and scores 0.250 on every pair. 37 of 43 pairs fall below that line (blue), 6 above it (red).

1 Introduction

The Predictive AI Evaluation Competition asks how likely an AI system is to answer each question (item) of an unseen benchmark correctly. We call a system under a given configuration a subject. Predictions are scored after the subject reveals 0, 1, 3, 7, 15 and 31 of its own answers, by the Brier area under that learning curve (ALC). Budgets 0 to 3 carry half of the weight, and that is where

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predictors lose most. With no answers yet, a predictor cannot tell a system that answers nearly everything from one that answers half.

Our idea is to ask a language model. It has read about most named AI systems, so it can say how often a given system will succeed before the system answers anything. We turn its estimate into a prior that the system's own answers then update. Our contributions:

- **The finding** (Section 4): a language model knows roughly how strong a named AI system is, and the knowledge comes from the system's identity.
- **The effect** (Sections 5 and 6): starting from its estimate helps most before the first answer, on the hidden benchmarks and on public ones untouched by development.
- **A reusable kit**: the predictor, a simulator of the competition protocol, every cached language-model output, and 1,157 accuracy forecasts (757 on the main public benchmarks, 400 on 44 newer ones) that other methods can be scored on.

Related work. The organisers' Bayesian Linguistic Evaluator also starts from a language model's estimate, but it asks for every target response separately and revises the estimate in the prompt. We ask once per subject and benchmark and update the estimate numerically. Language models have also been asked to forecast benchmark scores from task descriptions [9], and missing scores can be completed from a model's other scores [10]. We use such a forecast as the prior of an item-level predictor. Methods that predict from few items usually assume items already calibrated on many models [1–4], which a new benchmark lacks. In measurement terms, the task is a Rasch model [5, 6] fitted online from very few responses, as in adaptive testing [7], and our prior is the collateral information that sets an examinee's starting ability [8].

2 Task

For each subject–benchmark pair, items are split 50/50 into an acquisition pool and an evaluation pool. Acquisition items stream past, and the program decides which to reveal, up to 31. Predictions on the evaluation pool are scored at budgets $b \in \{0, 1, 3, 7, 15, 31\}$ with weights $0.1, 0.2, 0.2, 0.2, 0.2, 0.1$, and the ALC is averaged over pairs. The predictor sees the labels revealed for every pair in a run. The program sees item text and subject attributes, never reference answers.

3 Method

Subject prior. We ask Claude Opus 5.5 what share of the benchmark's items the subject answers correctly, given up to eight sample items and the subject's name, provider, release date, harness and reasoning effort. Eight solved examples from public benchmarks precede the question and fix the scale. The estimate q_s is shrunk toward one half, $m_s = 0.5 + 0.75\,(q_s - 0.5)$, and acts as a Beta prior with four pseudo-counts:

$$\hat{\mu}_s = \frac{\sum_{j \in R_s} y_{sj} + 4\, m_s}{|R_s| + 4}, \tag{1}$$

where R_s is the set of items s has revealed.

The rest of the predictor. When other subjects share the benchmark, $\hat{\mu}_s$ is also pulled toward their success rate. Each item's prediction then moves with three signals: the answers other subjects gave to the same item (one Newton step of a Rasch model), the residuals of similar revealed items, and a difficulty rating from Claude Opus calibrated on the revealed labels. The acquisition rule reveals items no other subject has revealed yet.

4 The language model knows the system

We measured the prior on public benchmarks, against each subject's true accuracy. We express its error as the number n of the subject's own answers whose Laplace estimate $(k+1)/(n+2)$ has the same mean squared error.

Table 1: Error of the estimated accuracy of a system on a benchmark, before the system answers anything. This is the RMSE of the accuracy estimate against the system's true accuracy, not the Brier score of Figures 1 and 2. Lower is better. The blocks are different samples, so compare rows only within a block. Full table in Appendix A.

Estimate	Error
277 pairs with an identifiable system, from 24 public benchmarks	
Guess 50%	0.197
Average accuracy of all systems on the other benchmarks	0.199
Language model (Opus 5.5, or Sonnet and Haiku where Opus gives no number)	**0.168**
400 pairs from 44 newer benchmarks that no part of our development touched	
Guess 50%	0.247
Language model (as above)	**0.133**
An earlier sample of 71 pairs	
Guess 50%	0.217
Opus 5.5	0.141
with the system's name, provider and date hidden	0.211
with the benchmark named in the prompt	0.136
Haiku 5.5	0.185

Table 2: The submitted predictor on 90 identical simulated draws. Differences are paired over draws ($\pm$ one standard error over draws), against the variant with both language-model parts off. Counting a pair's remaining labels out of 31, as the platform does, gives the same ALC values.

Variant	ALC	B_0	Difference	Better in
Laplace estimate on own answers	0.2020	0.250	$+0.0008 \pm 0.0002$	20% of draws
Neither part	0.2012	0.250	–	–
Item ratings only	0.1992	0.244	-0.0020 ± 0.0004	68% of draws
Subject prior only	0.1941	0.221	-0.0071 ± 0.0009	80% of draws
Both (submitted)	**0.1924**	0.216	-0.0088 ± 0.0010	83% of draws

Table 1 shows the result. On the 277 named pairs from the 24 public benchmarks that supply no calibration example, the deployed prior is worth four of the subject's own answers (90% interval 0 to 10, bootstrap over benchmarks). Priors from data alone do not work before the first answer: the average benchmark's accuracy is worth nothing on these pairs, and so is a subject's level fitted on its other public benchmarks.

Where the knowledge comes from. The bottom rows of Table 1 answer this. When we hide the subject's name, provider and release date, the estimate falls almost to the constant, so the value comes from knowing which system it is. When we name the benchmark, which the deployed prompt never does, the estimate barely changes, so the language model is not retrieving a stored score for this benchmark. It also cannot know scores on the competition's hidden benchmarks, which it has never seen. A stronger language model knows more: Haiku is worth three answers, Sonnet and Opus seven. What it lacks is how hard the benchmark is, which is why a calibration example on the same benchmark raises its worth to twelve answers. A check on systems released after the language model's training cutoff is in Section 7.

5 The prior improves the predictor

To isolate the prior, we ran the submitted predictor on identical simulated draws and switched its language-model parts off in turn. Each draw holds 3 to 8 subject–benchmark pairs (5.7 on average), mostly one subject per benchmark (Appendix B). The subject priors are real Opus calls with the deployed prompt, cached so every variant sees the same estimates.

Table 2 gives the result, and Figure 2 sets it against the true accuracy. With one subject per benchmark, the predictor without its language-model parts is barely better than a smoothed average of the

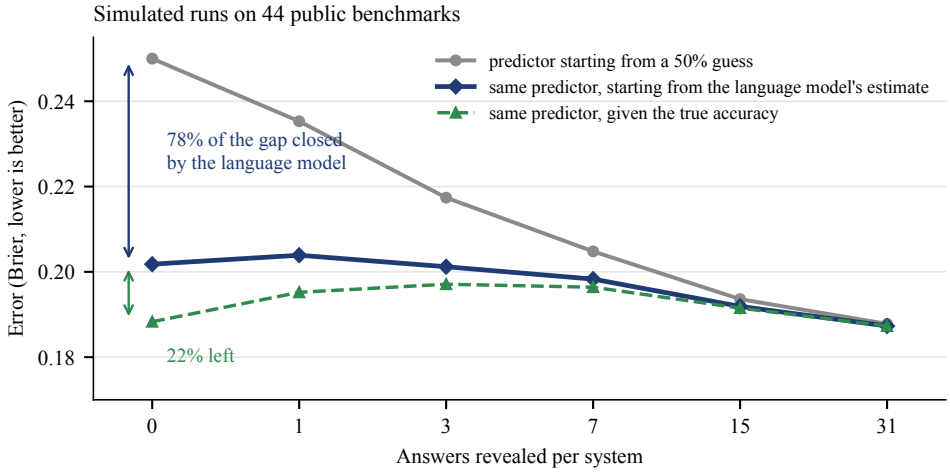


Figure 2: The submitted predictor on 30 simulated draws over 44 public benchmarks that no part of our development touched (149 pairs with an identifiable system), started from a 50% guess, from the language model's estimate, and from each system's true accuracy, all through the same shrinkage toward one half. Before the first answer the language model closes 78% of the gap, and 73% of the gap to the true accuracy used without shrinkage. With a good starting point the error rises over the first few answers, because the predictor lets them move it too far.

subject's own answers. The prior is the largest part of the gain. It lowers the error at budget 0 from 0.250 to 0.221, and its effect fades by budget 7, once the subject's own answers take over.

What else we tried. We also tried to get more out of the item ratings and to learn the whole predictor from data. A fine-tuned difficulty model, a per-benchmark choice between rating sources, and one 4B network that reads every revealed answer gave no reliable improvement (Appendix D). The difficulty model's small gain came from a single safety benchmark, and it lost on three benchmarks it had never seen.

6 Live results

Each formative run draws nine subject–benchmark pairs from seven benchmarks, and in our runs most benchmarks had a single subject. Raw scores swing with the draw. Three runs of the same final package scored 0.130, 0.181 and 0.164. Budget 0 still allows a paired comparison. The pooled baseline predicts one half for every item before any answers, so its Brier score at budget 0 is 0.250 on any draw. With the prior the score was 0.193 on the 43 pairs of the five runs with the prior (90% interval 0.159 to 0.227, bootstrap over the seven hidden benchmarks), lower on 37 of them. These are the five runs with the prior and without item ratings. The three runs of the final package, which add Opus item ratings, scored 0.187 on their 26 pairs. It helped on average on six of the seven hidden benchmarks. On the seventh it made the error worse ($+0.018$, on three of its five pairs). The feedback identifies benchmarks only by an anonymous code, so we cannot say which benchmark that is or why. Later budgets offer no such paired comparison. Appendix C lists every run.

7 Limitations

The prior depends on the language model knowing the subject. For unnamed subjects the predictor leaves it out. In simulation on 15 benchmarks never used in development it made no measurable difference: -0.0011 on those with named subjects (90% interval -0.009 to $+0.007$), helping on five and hurting on three, in each case where its starting estimate was closer or further than one half (Appendix B). On 44 newer benchmarks, also untouched by development and with identifiable systems, it lowered the ALC from 0.214 to 0.198 (Figure 2). The simulation benchmarks also served for development, so Table 2 may be optimistic. Online subjects were further from one half than those in the training data, and the shrinkage toward one half makes the prior too cautious for

them. The live comparison spans different draws, and we did not compare against the organisers' Bayesian Linguistic Evaluator, whose default configuration failed on the current public data. We submitted a fix (paiec_baseline#4). The method needs a paid API key, and all benchmarks we used have binary responses. Most subjects were released before the language model's training cutoff. On three systems released after it, on one public agent benchmark, the prior beat guessing, but its advantage over the average of earlier systems came from one of the three.

8 Reproducibility and disclosures

Code, the simulator, cached language-model outputs and the logs behind every simulation are at `https://github.com/sunnyadn/paiec-pooled-priors`. `python reproduce.py` prints the simulation tables from the logs. With access to the gated measurement-db, the documented scripts rebuild the item files, after which `reproduce.py` recomputes Table 1 and `reproduce.py -rerun` replays the simulated draws from stored identifiers and reruns the simulations without an API key. The final Codabench submission is 1017682. The submission is two files that need only the Python standard library and NumPy. Because the platform recreates its worker processes at every budget, the predictor stores every language-model result on disk for all workers to share and caps calls per worker. Without an API key it runs on labels alone.

Disclosures. Training data: public measurement-db only, for development, simulation and the eight calibration examples. Pretrained models: Claude Opus 5.5 through the Anthropic API at test time, with Claude Sonnet 5.5 and Claude Haiku 5.5 as fallbacks. During development we also compared GPT and Gemini models. Development used LLM coding assistants (Claude Code and Codex).

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A Prior quality in detail

Benchmark sets. The paper uses three sets of public benchmarks. The main evaluation has 32 benchmarks with named subjects, eight of which supply a calibration example, leaving the 24 of Table 1. Appendix B simulates 15 benchmarks used in no development step, eight of them with

Table 3: How well a language model estimates a subject's accuracy before any answers. The top rows use every named pair from the 24 public benchmarks that supply no calibration example, with the shrinkage fixed before this evaluation. The lower rows compare models on an earlier sample of 71 pairs. "Answers" is the number of the subject's own answers with the same mean squared error. *Uses other subjects' answers on the same benchmark, which do not exist before the first answer.

Estimate	Pairs	RMSE	Correlation	Answers
Constant 0.5	277	0.197	–	0
Mean accuracy of public benchmarks, this one left out	277	0.199	–	0
Subject's level from its other public benchmarks	277	0.203	–	0
Deployed: Opus, then Sonnet, then Haiku	277	**0.168**	0.55	**4**
Subject's level plus this benchmark's level from other subjects*	277	0.143	–	7
Constant 0.5	71	0.217	–	0
Claude Haiku 5.5	71	0.185	0.63	3
Claude Sonnet 5.5	71	0.142	0.76	7
Claude Opus 5.5	71	0.141	0.76	7
Opus, subject's identity hidden	71	0.211	0.28	1
Opus, benchmark named in the prompt	71	0.136	0.78	8

mostly named subjects. Figure 2 and the third block of Table 1 use 44 benchmarks added to the public data after development, with identifiable systems. Their subjects are further from one half (guessing scores 0.247 there against 0.197 on the main set), and the prior's error is about half that of guessing there, against 85% of it on the main set.

The full evaluation takes every public pair with a named subject and at least 50 responses, one subject per name and benchmark: 765 pairs on 32 benchmarks, of which the eight calibration pairs are left out. Every estimate saw five sample items (the deployed prompt shows up to eight), and the shrinkage of 0.75 was fixed before this evaluation. Opus gave a number on 678 pairs. On the other 79, all from `alignment_faking` and `harmbench`, the deployed predictor falls back to Sonnet and then Haiku. Table 4 splits the result. The data-only priors in Table 1 are fitted on the public response tables with the target pair left out. The subject's level and the benchmark's level are additive on the logit scale, fitted by alternating ridge regression; a subject seen on no other public benchmark (127 of the 277 pairs) gets level zero. The lower rows of Table 1 come from an earlier sample of at most four pairs per benchmark, using the 71 on which all three models gave a number.

Table 4: The deployed prior on every named public pair, with 90% intervals from a bootstrap over benchmarks.

Benchmarks	Pairs	RMSE	Correlation	Answers
All benchmarks	757	0.139 [0.121, 0.171]	0.67	8 [4, 11]
Benchmarks with a calibration example	480	0.119 [0.111, 0.129]	0.74	12 [9, 13]
Benchmarks without one	277	0.168 [0.128, 0.201]	0.55	4 [0, 10]
of which never used in development	32	0.177 [0.134, 0.215]	0.71	4 [1, 8]

B Benchmarks never used in development

The simulator of Section 5 samples seven benchmarks per draw with up to two subjects each and keeps a benchmark only if enough subjects answered 70% of its 120 sampled items. We repeated the paired comparison on 15 benchmarks used in no development step, with item ratings off in both arms. Seven have mostly unnamed subjects, where the predictor leaves the prior out and the difference is zero by design. On the eight with mostly named subjects the prior changed the ALC by -0.0011 averaged over pairs (90% interval -0.009 to $+0.007$ from a bootstrap over benchmarks). Table 5 splits the change by budget. On every benchmark the change in ALC follows the change at budget 0: the prior helps where its starting estimate is closer than one half and hurts where it is further, and either way its effect is nearly gone by seven answers.

Table 5: The subject prior on the eight held-out benchmarks with named subjects: change in Brier score at budgets 0, 1, 3 and 7 and in ALC, prior minus no prior, over the same 90 draws (negative is better).

Benchmark	Pairs	B_0	B_1	B_3	B_7	ALC
`bertaqa`	27	-0.059	-0.041	-0.023	-0.009	-0.021
`evouna`	54	-0.050	-0.036	-0.012	-0.002	-0.015
`aider`	49	-0.029	-0.025	-0.018	-0.004	-0.013
`healthadminbench`	47	-0.035	-0.018	-0.006	-0.000	-0.008
`edu_circuit_hw`	46	-0.015	-0.010	-0.007	-0.003	-0.005
`clasheval`	51	$+0.017$	$+0.009$	$+0.006$	$+0.000$	$+0.005$
`dbpa`	59	$+0.051$	$+0.027$	$+0.013$	$+0.003$	$+0.014$
`ehrflowbench`	51	$+0.056$	$+0.035$	$+0.025$	$+0.012$	$+0.023$
All eight	384	-0.003				-0.001

Table 6: Every online formative run. Section 6 uses the five runs marked "+ subject prior". B_{31} is the mean Brier score at budget 31, and excess is the ALC above each pair's budget-31 level. The runs drew different subjects, so the rows are not a paired comparison.

Submission	Version	Pairs	Raw ALC	B_{31}	Excess
1016627	v6	9	0.158	0.116	0.042
1016628	v6	9	0.170	0.143	0.027
1016651	v6, no API key	8	0.158	0.134	0.024
1017498	+ subject prior (v19)	9	0.169	0.164	0.006
1017499	+ subject prior (v19)	9	0.159	0.151	0.008
1017545	+ subject prior (v23)	8	0.195	0.183	0.011
1017562	+ subject prior (v25)	8	0.179	0.169	0.010
1017629	+ subject prior (v27)	9	0.184	0.167	0.017
1017607	final	8	0.181	0.175	0.006
1017628	final	9	0.130	0.126	0.004
1017682	final (selected)	9	0.164	0.149	0.014

C Online runs

D Variants that did not improve the submission

Table 7: Variants tried and dropped. Differences in pair-averaged Brier ALC against the predictor of the same stage (negative is better). Item ratings help only where a benchmark's labels move with ability. On some safety benchmarks a label of 1 means that an attack succeeded, which reverses the rating's direction, and in our tests nothing in a benchmark's description told the two kinds apart.

Variant	Result
Opus and GPT prior averaged (simulation)	-0.0011 ± 0.0003
same, online (4 runs)	B_0 excess 0.048–0.085 vs 0.031–0.049
Text-learned difficulty (n-grams, embeddings, LoRA)	correlation ≤ 0.08
Prior ensembles (Opus twice, Opus with Sonnet)	RMSE 0.139–0.142 vs 0.141
Gemini added to the Opus and GPT prior	RMSE 0.137 vs 0.138
Later predictor, four held-out benchmarks of our development split	
Fine-tuned 4B difficulty model instead of Opus ratings	-0.0012 (one refusal benchmark)
same, three benchmarks it never saw	$+0.0032$
Two rating slopes, chosen per benchmark from the labels	-0.0003
Ratings off where a language model judges label 1 an attack	-0.0021 held out, $+0.0021$ training
One 4B network reading all revealed answers (126,556 sequences)	$+0.0085$ (500 draws)